unmanned aviation. Conceived during World War I, it was designed as an unmanned aerial torpedo capable of carrying an explosive payload to a predetermined target. The aircraft was built by the US Army Signal Corps in collaboration with the Dayton-Wright Airplane Company, marking one of the earliest large-scale military efforts to create an automated flying weapon.

The Kettering Bug operated on a mechanically pre-set guidance system. Instead of real-time radio control, it used a simple yet ingenious mechanical timer that counted engine revolutions to estimate distance traveled. Once the pre-set number of revolutions was reached, the engine would shut off, and the wings would detach, allowing the drone to glide toward the target with its 200-pound explosive payload. This innovation represented the first known attempt to automate flight and payload delivery, introducing the fundamental concept of pre-programed flight control that would later evolve into modern autopilot and missile guidance systems.

Constructed primarily from wood, cardboard, and fabric, the Kettering Bug had a wingspan of about 12 feet (3.7 meters) and was powered by a 40-horsepower, four-cylinder Ford engine, giving it a range of approximately 75 miles (120 kilometers). Although it was never used in active combat before the end of World War I, its successful test flights demonstrated that unmanned aircraft could accurately follow a pre-determined course without human intervention.

The significance of the Kettering Bug lies in its role as a technological bridge between early experimental UAVs and modern cruise missiles. It introduced the concept of autonomous flight based on mechanical programing, influencing later developments in aerial weapons, navigation systems, and automatic control theory. In essence, the Kettering Bug was a pioneering precursor to today’s precision-guided munitions, illustrating how